

Mapping the UK Climate & Nature Policy Landscape for a new UK daily sunshine-duration product

Gap assessment · Prepared July 2026

Key points

- **This is a gap assessment of a *daily* product, not of sunshine data in general.** The Met Office already produces gridded UK sunshine at 1 km, but only at *monthly* and coarser resolution in HadUK-Grid (sunshine series from 1910); of the HadUK-Grid variables, only rainfall and max/min air temperature are gridded *daily*. A near-real-time, spatially complete **daily** sunshine-duration product does not yet exist as an operational, policy-facing dataset. That is the gap.
- **The method already exists in-house, and a peer service already ships the product.** Good (2010) demonstrated Met Office derivation of gridded *daily* sunshine over the UK from Meteosat/SEVIRI cloud retrievals; Germany's DWD already operationalises blended station–satellite gridded solar-radiation products, including a real-time one (Pfeifroth et al. 2026). A prospective UK product would operationalise the Good method — satellite-blended, daily, ~1 km, national, near-real-time — plus derived surface-solar-resource layers.
- **Highest-value applications are in energy.** Solar output is directly and mechanistically driven by sunshine/irradiance (Harrison 2023). A daily product would support Clean Power 2030 solar delivery (45–47 GW by 2030), the Solar Roadmap, generation forecasting and system balancing — a channel already live through the **Met Office–NESO** collaboration.
- **Most mature** applications are solar delivery and system integration (Demonstrated — sunshine/irradiance resource data is already used operationally in the energy sector, and shipped as a product by DWD); **least mature** are the biodiversity links (phenology, pollinators), which are Conceptual for direct policy use.
- **Caveat:** the supplied bibliography is an external sunshine reading list, so Met Office capability was scoped from Met Office / CEDA sources. Where a policy link is inferred rather than documented it is marked the lower maturity value (Conceptual) and flagged.

The evidence base for a prospective UK daily sunshine-duration product

A prospective product would deliver **daily** satellite-blended sunshine-duration fields for the UK at ~1 km, near-real-time, with derived surface-solar-resource layers — closing the gap between the existing HadUK-Grid *monthly* sunshine grids and the *daily* cadence that energy, agriculture and biodiversity applications need. The Met Office already holds every component: the HadUK-Grid gridding pipeline (National Climate Information Centre), the Good (2010) satellite-derived *daily* sunshine method, State-of-the-UK-Climate monitoring, and an operational energy-services channel (Met Office–NESO). The external evidence base below shows the policy-relevant applications are established science.

- **Solar energy conversion** — daily sunshine duration (bright hours) converts directly to daily solar power output, structured by weather type (Harrison 2023, 1908–2021 records).
- **National met-service precedent** — Germany's DWD already operationalises gridded surface-solar-radiation products blending station and satellite data, both climate-focused (EUMETSAT CM SAF) and real-time ('DUETT'), all publicly available (Pfeifroth et al. 2026) — a direct precedent for the prospective UK product.
- **Societal value** — increased UK sunshine is linked to solar generation capacity, tourism and human well-being (Spellman 2024).
- **Temperature coupling** — daily sunshine duration is strongly and significantly correlated with daily maximum temperature across European/UK stations (van den Besselaar et al. 2015), relevant to heat, growing-season and agri-productivity signals.
- **Record continuity** — homogenisation between Campbell–Stokes and automated Kipp & Zonen sensors (Legg 2014; Kerr & Tabony 2004) is the data-quality basis any operational daily product must carry through.

A daily product would translate these demonstrated relationships into a national, policy-ready evidence layer for solar delivery, climate-risk monitoring and — more tentatively — agri-environment and biodiversity attribution; the table maps where that would matter and how proven each use is.

Policy applications, relevance and maturity

Two independent axes. **Relevance** (materiality of a daily sunshine-duration product to the policy): **3 High** = would be a direct, central input; **2 Medium** = a useful supporting evidence layer; **1 Low** = marginal or indirect. **Maturity** (how proven the application is): Demonstrated = proven in operational or industry use and transferable; Emerging = partial evidence, needs development/validation for this use; Conceptual = plausible but unproven. Items marked † are activities not held in the governance workbook. Policy names link to the source page.

Ref	Policy activity	Relevance	Maturity	What a daily sunshine-duration product could enable for this policy
POL-005	Clean Power 2030 Action Plan	3 HIGH	Demonstrated	Central to the plan is 45–47 GW of solar by 2030. Solar generation is directly weather-dependent (solar irradiance sets output). A daily, spatially complete sunshine-duration product would sharpen resource assessment, capacity-factor assumptions, generation forecasting and system integration — a use already demonstrated operationally through the Met Office–NESO energy-services collaboration.
†	Solar Roadmap 2025 (DESNZ / Solar Taskforce / Solar Council)	3 HIGH	Demonstrated	Published 30 June 2025; 72 actions to reach 45–47 GW, including siting, rooftop and ground-mount deployment, and agrivoltaics. Solar-resource information underpins yield estimation and well-sited deployment; industry already uses irradiance/sunshine climatologies for this, so an authoritative national daily product is a Demonstrated, transferable input. Not held as a workbook record.
POL-004	UK Net Zero Strategy / Carbon Budget Delivery Plan (CBDP) 2023	2 MED	Emerging	Electricity-sector decarbonisation pathways rest on solar-generation projections that embed resource assumptions. A consistent daily sunshine record would improve those assumptions and the monitoring of realised solar output against the pathway — evidenced in principle, not yet an explicit CBDP input.
POL-016	Contracts for Difference (CfD) scheme	2 MED	Emerging	Over 7 GW of solar has been contracted through CfD. Strike-price setting and generation/revenue modelling for solar depend on the underlying resource; a national daily sunshine climatology would provide a consistent, official basis for yield assumptions. Currently supplied by commercial resource data rather than an MO product.
POL-050	National Planning Policy Framework (NPPF) — solar-farm siting	2 MED	Emerging	Ground-mount solar potential is markedly higher in southern England; planning for solar farms (and the raised NSIP threshold) weighs resource against agricultural-land and landscape impacts. A national daily/aggregated sunshine-resource layer would give plan-makers a consistent siting evidence base — used informally today, not codified.
†	Future Homes Standard (rooftop solar on new homes, from 2027)	2 MED	Emerging	From 2027 most new homes in England must include rooftop solar. Location-specific sunshine climatology informs realistic yield and bill-saving estimates that underpin the standard's impact assessment and household guidance. Evidenced in principle; not currently an MO product feeding the standard. Not held as a workbook record.
POL-017	Climate Change Risk Assessment (CCRA3 2022; CCRA4 from 2026–27)	2 MED	Emerging	Solar-resource variability (energy-system risk), growing-season and agri-productivity signals, and well-being/heat all bear on CCRA risks and opportunities. Sunshine is a monitored State-of-Climate variable but not yet a discrete CCRA evidence layer; a daily product is a credible CCRA4 input.
POL-002	National Adaptation Programme (NAP3) 2023–2028	2 MED	Emerging	NAP delivers CCRA risks, including agriculture and energy-system resilience. Spatially explicit daily sunshine would support solar-resilience and crop-productivity actions. The link is inferred from NAP themes rather than documented, so scored the lower maturity value.
POL-013	Environmental Land Management (ELM) Schemes	1 LOW	Conceptual	Solar radiation / sunshine drives crop and grassland growth and is relevant to agrivoltaics (dual land use flagged in the Solar Roadmap). A daily product could inform scheme targeting and dual-use siting, but this is an inferred, not-yet-articulated pathway.
POL-003	UK National Biodiversity Strategy and Action Plan (NBSAP) 2025	1 LOW	Conceptual	Sunshine drives phenology and the flight activity of pollinators and butterflies that underpin several biodiversity indicators. A daily sunshine product could improve attribution of biodiversity change to climate and correct survey-condition effects — credible but unproven for direct NBSAP use; inferred link.

Policy indicators

A scan of *indicators_climate_nature.xlsx* for indicators whose measured quantity a daily sunshine-duration product could directly inform — solar-driven generation, solar-radiation-driven crop yield, and sunshine-dependent phenology and insect flight activity. Deployment counts whose measured quantity takes no climate input (e.g. installed solar PV capacity in GW) were excluded on the same basis as GIS-area indicators. Framework and application columns link to the specific report / indicator page where each is defined or used (drawn from the workbook's Source_URL column where present). Maturity for the biodiversity indicators is Conceptual for direct policy use; the energy and agriculture indicators are the more proven links.

Indicator (ref)	Indicator name	Source framework	Current policy application(s)
IND-C-056	Energy from offshore wind and solar as % of electricity mix (GB)	CCC Mitigation Monitoring Framework (electricity-supply key indicators)	Clean Power 2030 Action Plan (45–47 GW solar by 2030) / CBDP — Climate Change Act 2008 carbon budgets
IND-C-110	Agricultural crop yield variability (wheat, oilseed rape) (UK) — sunshine informs the solar-radiation term, not the whole outcome	CCC Adaptation Monitoring Framework 2023–25	NAP3 (food-security & agriculture resilience) / CCRA — Climate Change Act 2008, s.58
IND-J-063	Spring Index (phenology) (UK)	JNCC UKBI 2025 — Spring Index	NBSAP 2025 (Environment Act 2021 targets); reported via the Environmental Indicator Framework (EIP progress report) and UK CBD/GBF reporting
IND-J-014	Insects of the wider countryside – all butterflies (UK) — sunshine drives the transect flight-activity term	JNCC UKBI 2025 — Insects of the wider countryside (butterflies)	NBSAP 2025 (Environment Act 2021 species targets); reported via the Environmental Indicator Framework (EIP progress report) and UK CBD/GBF reporting
IND-J-045	Pollinating insects (UK) — weather-dependent survey counts	JNCC UKBI 2025 — Pollinating insects	NBSAP 2025 (Environment Act 2021 species targets); reported via the Environmental Indicator Framework (EIP progress report) and UK CBD/GBF reporting

Methods

- Capability scoped from external evidence and Met Office sources**, not from an MO output list: the supplied bibliography is an external UK sunshine reading list, used to establish which sunshine–policy relationships are demonstrated (solar conversion, temperature coupling, societal value) and how mature each is — including a peer national-met-service precedent (DWD gridded solar-radiation products, Pfeifroth et al. 2026).
- Met Office capability characterised** by inspecting Met Office and CEDA pages (accessed July 2026): HadUK-Grid (1 km gridded sunshine, *monthly* only — daily grids exist for rainfall and temperature but not sunshine); the Good (2010) satellite-derived *daily* sunshine method; State of the UK Climate monitoring (National Climate Information Centre); and the operational Met Office–NESO energy-services channel.
- Policy candidates** drawn from the governance workbook's Policies & Activities sheet (*uk_climate_nature_governance.xlsx*, canonical; POL IDs stable across versions), filtered on Climate, Energy, Land Use & Agriculture and Nature sectors, then screened for a plausible dependency on daily sunshine / surface-solar resource; supplemented with solar activities absent from the workbook (Solar Roadmap 2025, Future Homes Standard), marked †.
- Scoring** used two independent axes. *Relevance* (materiality of a daily sunshine product): central input = 3; supporting evidence layer = 2; marginal/indirect = 1. *Maturity*: Demonstrated where the application is already used operationally or in industry and transferable; Emerging where partial evidence needs development for the specific policy use; Conceptual where the use is plausible but unproven. Kept separate so a high-value-but-immature application is not mistaken for a low-relevance one.
- Indicator scan** of *indicators_climate_nature.xlsx* across its framework sheets, keyword-filtered for sunshine, solar, radiation, phenology, growing season and crop yield, then narrowed to indicators whose measured quantity a daily sunshine product could directly inform (solar generation share, crop-yield variability, phenology and insect flight activity); installed-capacity (GW) indicators were excluded because their measured quantity takes no climate input. Framework/application links use the workbook's Source_URL column where present. Sunshine duration was treated as distinct from, though correlated with, surface solar irradiance and air temperature.

Main gaps and caveats

Two framing caveats. First, this assesses a prospective *daily* product: the Met Office already supplies gridded *monthly* sunshine (HadUK-Grid) and the satellite *daily* method exists (Good 2010), and a peer service (DWD) already ships an operational blended product — so the gap is operationalisation and cadence, not first-principles capability; several energy links are therefore

Demonstrated rather than Conceptual. Second, sunshine duration, surface solar irradiance and air temperature are related but distinct; a policy-ready product would need to state which quantity it delivers (bright-sunshine hours vs. irradiance) because energy applications ultimately want irradiance/PV yield, for which sunshine duration is a strong but not exact proxy — the DWD experience of moving from sunshine duration to solar-radiation products is instructive here. The governance workbook holds no records for the Solar Roadmap 2025 or the Future Homes Standard — the two highest-relevance solar activities — arguably worth adding. The ELM and NBSAP links are inferred (Conceptual): sunshine matters mechanistically but is not named in current scheme or strategy text. The Future Homes Standard URL points to the consultation page; the final standard's publication URL should be substituted once confirmed.

References

Evidence base (external literature)

6. Spellman, G. (2024) Sunnier climes in the UK. *Weather* 79(1): 27–31.
7. Good, E.J. (2010) Estimating daily sunshine duration over the UK from geostationary satellite data. *Weather* 65(12): 324–328. (Met Office — satellite-derived daily sunshine method.)
8. Harrison, R.G. (2023) Sunshine and solar power in the UK. *Theoretical and Applied Climatology* 155: 1163–1174.
9. van den Besselaar, E.J.M. et al. (2015) Relationship between sunshine duration and temperature trends across Europe. *Journal of Geophysical Research: Atmospheres* 120(20): 10,730–10,739.
10. Legg, T.P. (2014) A review of progress in establishing a consistent UK sunshine series. *Met Office Hadley Centre Climate Memorandum*.
11. Kerr, A. & Tabony, R.C. (2004) Comparison of sunshine recorded by Campbell–Stokes and automatic sensors. *Weather* (Royal Meteorological Society).
12. Pfeifroth, U., Trentmann, J., Brinckmann, S. & Hollmann, R. (2026) An overview of surface solar radiation data products at the German meteorological service (DWD). *Journal of the European Meteorological Society* (Elsevier) — EUMETSAT CM SAF climate records and the real-time station–satellite 'DUETT' product; direct precedent for an operational blended UK product.

Met Office capability and other sources

- Met Office / CEDA, HadUK-Grid gridded climate observations (v1.3.2.ceda, June 2026; sunshine gridded monthly from 1910, at 1 km). metoffice.gov.uk/hadobs/hadukgrid
- Met Office, HadUK-Grid FAQ — sunshine recorder types (Campbell–Stokes vs Kipp & Zonen) and grid variables (accessed July 2026).
- Kendon, M. et al. (2025) State of the UK Climate in 2024. *International Journal of Climatology* 45(S1): e70010 — UK sunshine 2015–2024 3% above 1991–2020, 8% above 1961–1990; winter/spring driven.
- Met Office (Jan 2026) 2025 confirmed as the UK's sunniest year on record (1648.5 hours); notes the National Energy System Operator (NESO)–Met Office collaboration and solar's growing share of GB supply.
- DESNZ (30 June 2025) Solar Roadmap: United Kingdom powered by solar. gov.uk/government/publications/solar-roadmap
- JNCC / Defra (2 Dec 2025) UK Biodiversity Indicators 2025 — Spring Index, butterflies and pollinating-insects indicator pages; domestic use via the Environmental Indicator Framework and UK CBD/GBF reporting. jncc.gov.uk/our-work/uk-biodiversity-indicators
- Climate Change Committee (2025) Progress in reducing emissions and Progress in adapting to climate change — 2025 reports to Parliament (mitigation and adaptation monitoring frameworks).
- `uk_climate_nature_governance.xlsx` (Policies & Activities) and `indicators_climate_nature.xlsx` (framework sheets) — canonical workbooks.